

§ 23. Transcendence Degree and the Division of Fields into Classes.

In the following considerations, in which as before we refer to a ground field $\mathfrak{K}$, the well-ordering theorem again enters into use.

1. *Every system $\mathfrak{S}$ whose elements are not all algebraic with respect to $\mathfrak{K}$ is equivalent to an irreducible subsystem.*

Proof. If $\mathfrak{S}$ contains algebraic elements, then after omitting them a system equivalent to $\mathfrak{S}$ remains, and we may continue with it. We may therefore assume from the outset that every element of $\mathfrak{S}$ is transcendental with respect to $\mathfrak{K}$. If first $\mathfrak{S}$ is finite, and $\mathfrak{T}$ is an equivalent subsystem of $\mathfrak{S}$ with as few elements as possible, then $\mathfrak{T}$ is plainly irreducible.

If the system $\mathfrak{S}$ is infinite, we regard it as well-ordered, $\mathfrak{S} = \{a_\nu\}$, and denote by $\mathfrak{S}_\nu$ the segment of $\mathfrak{S}$ determined by a_ν . We omit from $\mathfrak{S}$ every element a_ν ($\nu > 0$) which is algebraically dependent on its associated segment $\mathfrak{S}_\nu$. There remains a system $\mathfrak{S}'$ which is irreducible and equivalent to $\mathfrak{S}$. First, a_0 , since it belongs to $\mathfrak{S}'$, is algebraically dependent on $\mathfrak{S}'$. Suppose further that it has already been proved that the elements of the segment $\mathfrak{S}_\nu$ are algebraically dependent on $\mathfrak{S}'$. Then the same is true of a_ν , because a_ν either is algebraically dependent on $\mathfrak{S}_\nu$ or belongs to $\mathfrak{S}'$. This gives the equivalence of $\mathfrak{S}$ and $\mathfrak{S}'$. If $\mathfrak{S}'$ were reducible, and if $\mathfrak{L}$ were a finite reducible subsystem of $\mathfrak{S}'$ with as few elements as possible, and a_ν the last element of $\mathfrak{L}$, then, since a_ν is transcendental with respect to $\mathfrak{K}$, it could not be the sole element of $\mathfrak{L}$; and the system $\mathfrak{L}'$ which remains after omitting a_ν from $\mathfrak{L}$, being part of $\mathfrak{S}_\nu$, would be irreducible. Hence, by § 22, 6, a_ν would be algebraically dependent on $\mathfrak{L}'$, and therefore on $\mathfrak{S}_\nu$. But then a_ν would not belong to $\mathfrak{S}'$, a contradiction. Thus $\mathfrak{S}'$ is irreducible, and Theorem 1 is proved.

Now let $\Omega = \mathfrak{K}(\mathfrak{S})$ denote an arbitrary transcendental extension of $\mathfrak{K}$. Then $\mathfrak{S}$ is equivalent to an irreducible system $\mathfrak{S}'$; hence the elements of $\mathfrak{S}$ are algebraic with respect to $\mathfrak{K}(\mathfrak{S}')$. Thus $\mathfrak{K}(\mathfrak{S})$ is an algebraic extension of $\mathfrak{K}(\mathfrak{S}')$, while $\mathfrak{K}(\mathfrak{S}')$ is a purely transcendental extension of $\mathfrak{K}$. We obtain:

2. *Every transcendental extension of a field can be decomposed into a purely transcendental extension followed by an algebraic extension.*

Of course such a decomposition can be made in infinitely many ways. Suppose that $\mathfrak{S}_1$ and $\mathfrak{S}_2$ are two irreducible systems in Ω , on which all elements of Ω are algebraically dependent. Then $\mathfrak{S}_1$ and $\mathfrak{S}_2$ are equivalent. If one of these systems is finite, it was shown above that the other is also finite and contains the same number of elements (§ 22, 9). We shall show that in the case of infinite systems as well the cardinal numbers agree, and hence, by § 22, 5, that $\mathfrak{K}(\mathfrak{S}_1)$ and $\mathfrak{K}(\mathfrak{S}_2)$ are equivalent extensions of $\mathfrak{K}$. For this purpose we first introduce some notation.

Let $\{c_\mu\}$ ($0 \leq \mu < \tau$) be a well-ordered sequence of elements of an extension field, let τ be a limit number, and suppose that from a certain μ onward all elements c_μ are equal, say equal to a . We then say that the sequence has a limit, and that this limit is a ; in symbols

$$\lim_{\mu=\tau} c_\mu = a.$$

Suppose there is a well-ordered double sequence $\{c_{\mu\nu}\}$ ($0 \leq \mu < \tau$, $0 \leq \nu < \sigma$), which we think of as ordered in rows and columns, and let the rows be denoted by $\mathfrak{W}_\mu$, so that $\mathfrak{W}_\mu = \{c_{\mu\nu}\}$ ($0 \leq \nu < \sigma$). If τ is a limit number and every column has a limit

$$\lim_{\mu=\tau} c_{\mu\nu} = c'_\nu \quad (0 \leq \nu < \sigma),$$

then we say that the sequence $\{\mathfrak{W}_\mu\}$ has the limit $\mathfrak{W}' = \{c'_\nu\}$.

Let $\mathfrak{U}$ and $\mathfrak{B} = \{v_\mu\}$ ($0 \leq \mu < \tau$) be two irreducible systems, well-ordered in arbitrary ways; suppose that $\mathfrak{B}$ is algebraically dependent on $\mathfrak{U}$. Let v_μ be an arbitrary element of $\mathfrak{B}$, and let

$\mathfrak{B}_\mu$ denote the segment of $\mathfrak{B}$ determined by v_μ . By § 22, 7, $\mathfrak{U}$ contains a definite least subsystem $\mathfrak{L}$ on which v_μ is algebraically dependent. The system $\mathfrak{L}$ may well contain elements that occur in $\mathfrak{B}_\mu$, but it certainly also contains elements that do not occur in $\mathfrak{B}_\mu$. If we replace the last of these by v_μ , then $\mathfrak{U}$ is transformed into an irreducible system, equivalent to $\mathfrak{U}$, with the same ordinal number. For this system, completely determined by $\mathfrak{U}$, $\mathfrak{B}$, and μ , we introduce the notation $\vartheta_\mu(\mathfrak{U}, \mathfrak{B})$.

Now let $\mathfrak{U} = \{u_\nu\}$ ($0 \leq \nu < \sigma$) and $\mathfrak{B} = \{v_\mu\}$ ($0 \leq \mu < \tau$) be irreducible well-ordered systems, and suppose that $\mathfrak{B}$ is algebraically dependent on $\mathfrak{U}$. Suppose further that there is a double sequence $\{c_{\mu\nu}\}$ ($0 \leq \mu \leq \tau$, $0 \leq \nu < \sigma$), whose rows we denote by $\mathfrak{W}_\mu$. We shall assume the following relations:

- 1) Each row $\mathfrak{W}_\mu = \{c_{\mu\nu}\}$ ($0 \leq \nu < \sigma$) is an irreducible system and is equivalent to $\mathfrak{U}$, so that $\mathfrak{B}$ is algebraically dependent on every system $\mathfrak{W}_\mu$.
- 2) $\mathfrak{W}_0 = \mathfrak{U}$, hence $c_{0\nu} = u_\nu$ ($0 \leq \nu < \sigma$).
- 3) If $0 < \mu \leq \tau$ and μ is not a limit number, then

$$\mathfrak{W}_\mu = \vartheta_{\mu-1}(\mathfrak{W}_{\mu-1}, \mathfrak{B}).$$

- 4) If $0 < \mu \leq \tau$ and μ is a limit number, then the sequence $\{\mathfrak{W}_\lambda\}$ ($0 \leq \lambda < \mu$) has a limit, and this limit is $\mathfrak{W}_\mu$.

When these conditions are fulfilled, we say that the system sequence $\{\mathfrak{W}_\mu\}$ ($0 \leq \mu \leq \tau$), or also the double sequence $\{c_{\mu\nu}\}$, belongs to the system pair $\mathfrak{U}, \mathfrak{B}$. The premise for this language is therefore always that $\mathfrak{U}$ and $\mathfrak{B}$ are well-ordered irreducible systems and that $\mathfrak{B}$ is algebraically dependent on $\mathfrak{U}$.

The conditions 1) through 4) determine every row by the preceding rows; and since $\mathfrak{W}_0$ is to be $\mathfrak{U}$, we first obtain:

a) *To a system pair $\mathfrak{U}, \mathfrak{B}$ there can belong at most one system sequence.* Moreover our conditions immediately give:

b) If the system sequence $\{\mathfrak{W}_\mu\}$ ($0 \leq \mu \leq \tau$) belongs to the system pair $\mathfrak{U}, \mathfrak{B}$, and if $0 < \rho < \tau$, while $\mathfrak{B}_\rho$ is the segment of $\mathfrak{B}$ determined by v_ρ , then the system sequence $\{\mathfrak{W}_\mu\}$ ($0 \leq \mu \leq \rho$) belongs to the system pair $\mathfrak{U}, \mathfrak{B}_\rho$.

Assume now that the system sequence $\{\mathfrak{W}_\mu\}$ belongs to $\mathfrak{U}, \mathfrak{B}$, and select a column $\{c_{\mu\nu}\}$ ($0 \leq \mu \leq \tau$) from the double sequence. If not all elements of the column are equal to the initial element $c_{0\nu} = u_\nu$, let $c_{\rho\nu}$ be the first one different from u_ν . Then ρ cannot be a limit number; for otherwise, by 4), $c_{\rho\nu}$ would be the limit of $\{c_{\mu\nu}\}$ ($0 \leq \mu < \rho$), and so would be equal to preceding elements of the column. Since ρ is not a limit number, by 3) $\mathfrak{W}_\rho = \vartheta_{\rho-1}(\mathfrak{W}_{\rho-1}, \mathfrak{B})$; and $\mathfrak{W}_\rho$ is obtained from $\mathfrak{W}_{\rho-1}$ by replacing some element of $\mathfrak{W}_{\rho-1}$ by $v_{\rho-1}$. Since $c_{\rho\nu}$ is different from $c_{\rho-1,\nu}$, it follows that $c_{\rho\nu} = v_{\rho-1}$. But then $c_{\mu\nu} = v_{\rho-1}$ for every $\mu \geq \rho$. For otherwise let μ be the least ordinal number $> \rho$ for which $c_{\mu\nu} \neq v_{\rho-1}$. As before, it follows that μ is not a limit number and that $\mathfrak{W}_\mu$ is obtained from $\mathfrak{W}_{\mu-1}$ by replacing $c_{\mu-1,\nu} = v_{\rho-1}$ by $c_{\mu\nu} = v_{\mu-1}$. But by the definition of ϑ , $\vartheta_{\mu-1}(\mathfrak{W}_{\mu-1}, \mathfrak{B})$ is obtained from $\mathfrak{W}_{\mu-1}$ by replacing a certain element of $\mathfrak{W}_{\mu-1}$ which does not belong to the segment of $\mathfrak{B}$ determined by $v_{\mu-1}$, by $v_{\mu-1}$. The element $v_{\rho-1}$, however, does belong to this segment. Hence:

c) If the system sequence $\{\mathfrak{W}_\mu\}$ ($0 \leq \mu \leq \tau$), with $\mathfrak{W}_\mu = \{c_{\mu\nu}\}$, belongs to the system pair $\mathfrak{U}, \mathfrak{B}$, then in a column $\{c_{\mu\nu}\}$ ($0 \leq \mu \leq \tau$) either all elements are equal to u_ν , or there is an ordinal number ρ , $0 < \rho \leq \tau$, not a limit number, such that $c_{\mu\nu} = u_\nu$ for $\mu < \rho$, and $c_{\mu\nu} = v_{\rho-1}$ for $\mu \geq \rho$.

Under the same assumptions, select any element $v_{\rho-1}$ from $\mathfrak{B}$. By 3) it occurs in $\mathfrak{W}_\rho$, say as $c_{\rho\nu}$; and, as we have seen, when μ runs through the ordinal numbers $> \rho$, it cannot be replaced by any $v_{\mu-1}$. Thus $v_{\rho-1}$ occurs in every system $\mathfrak{W}_\mu$ whose index satisfies $\mu \geq \rho$. Therefore:

d) If the system sequence $\{\mathfrak{W}_\mu\}$ ($0 \leq \mu \leq \tau$) belongs to the system pair $\mathfrak{U}, \mathfrak{B} = \{v_\mu\}$ ($0 \leq \mu < \tau$), and if $\mathfrak{B}_\mu$ ($0 < \mu < \tau$) denotes the segment determined by v_μ , while $\mathfrak{B} = \mathfrak{B}_\tau$, then $\mathfrak{W}_\mu$ ($0 < \mu \leq \tau$) contains all elements of $\mathfrak{B}_\mu$ (of course not necessarily in the same order as $\mathfrak{B}_\mu$).

We now assert:

e) *To every pair $\mathfrak{U}, \mathfrak{B}$ of irreducible well-ordered systems, the second of which is algebraically dependent on the first, there belongs a system sequence.*

As before, put $\mathfrak{U} = \{u_\nu\}$ ($0 \leq \nu < \sigma$), $\mathfrak{B} = \mathfrak{B}_\tau = \{v_\mu\}$ ($0 \leq \mu < \tau$), and denote by $\mathfrak{B}_\mu$ the segment of $\mathfrak{B}$ determined by v_μ . In the case $\tau = 1$ we plainly obtain the required system sequence by putting $\mathfrak{W}_0 = \mathfrak{U}$, $\mathfrak{W}_1 = \vartheta_0(\mathfrak{U}, \mathfrak{B})$. For $\tau > 1$, e) is to be proved by induction. If τ is not a limit number, then to begin with there belongs to the system pair $\mathfrak{U}, \mathfrak{B}_{\tau-1}$ a system sequence $\{\mathfrak{W}_\mu\}$ ($0 \leq \mu \leq \tau - 1$); and if we put $\vartheta_{\tau-1}(\mathfrak{W}_{\tau-1}, \mathfrak{B}) = \mathfrak{W}_\tau$, it follows at once that the system sequence $\{\mathfrak{W}_\mu\}$ ($0 \leq \mu \leq \tau$) satisfies all conditions 1) through 4), and hence belongs to $\mathfrak{U}, \mathfrak{B}$.

Let τ be a limit number. If $0 < \lambda \leq \rho < \tau$, then to the pair $\mathfrak{U}, \mathfrak{B}_\rho$ there belongs one, and by a) only one, system sequence $\{\mathfrak{W}_\mu\}$ ($0 \leq \mu \leq \rho$); and by b) the subsequence $\{\mathfrak{W}_\mu\}$ ($0 \leq \mu \leq \lambda$) belongs to $\mathfrak{U}, \mathfrak{B}_\lambda$. Hence for each $\mu < \tau$ there exists a uniquely determined system $\mathfrak{W}_\mu$ such that the segment of the sequence $\{\mathfrak{W}_\mu\}$ ($0 \leq \mu < \tau$) extending through some $\mathfrak{W}_\rho$ inclusive is the system sequence belonging to $\mathfrak{U}, \mathfrak{B}_\rho$. We put $\mathfrak{W}_\mu = \{c_{\mu\nu}\}$ and consider a column $\{c_{\mu\nu}\}$ ($0 \leq \mu < \tau$). If not all elements are equal to u_ν , and if $c_{\rho\nu}$ is the first one different from u_ν , then all elements of the column following $c_{\rho\nu}$ are equal to $c_{\rho\nu}$; for the assumption $c_{\rho\nu} \neq c_{\chi\nu}$, $\rho < \chi$, is incompatible, by c), with the fact that the sequence $\{\mathfrak{W}_\mu\}$ ($0 \leq \mu \leq \chi$) belongs to $\mathfrak{U}, \mathfrak{B}_\chi$. Thus every column, and hence also the sequence $\{\mathfrak{W}_\mu\}$ ($0 \leq \mu < \tau$), has a limit. We put

$$\lim_{\mu=\tau} \mathfrak{W}_\mu = \mathfrak{W}_\tau, \quad \mathfrak{W}_\tau = \{c_{\tau\nu}\} \quad (0 \leq \nu < \sigma)$$

and assert that the system sequence $\{\mathfrak{W}_\mu\}$ ($0 \leq \mu \leq \tau$) belongs to $\mathfrak{U}, \mathfrak{B}$. To prove this, we have to show two things:

f) $\mathfrak{W}_\tau$ is irreducible,

g) $\mathfrak{W}_\tau$ is equivalent to $\mathfrak{W}_0 = \mathfrak{U}$.

If the system $\mathfrak{W}_\tau$ were reducible, it would contain a finite reducible subsystem $\mathfrak{L}$. Each element of $\mathfrak{L}$, being the limit of its column, is equal to preceding elements of that column; and since $\mathfrak{L}$ contains only finitely many elements, one can choose $\rho < \tau$ so that all elements of $\mathfrak{L}$ already occur in $\mathfrak{W}_\rho$. But then $\mathfrak{W}_\rho$ would be reducible, contrary to the fact that the system sequence $\{\mathfrak{W}_\mu\}$ ($0 \leq \mu \leq \rho$) belongs to $\mathfrak{U}, \mathfrak{B}_\rho$. This proves f).

An arbitrary element of $\mathfrak{B}$ may be denoted by $v_{\rho-1}$, where $\rho < \tau$. The element $v_{\rho-1}$ occurs in $\mathfrak{W}_\rho$, say as $c_{\rho\nu}$; and then in $\{c_{\mu\nu}\}$ ($0 \leq \mu < \tau$), for every $\mu \geq \rho$, one has $c_{\mu\nu} = v_{\rho-1}$, hence also $c_{\tau\nu} = \lim_{\mu=\tau} c_{\mu\nu} = v_{\rho-1}$. Thus $\mathfrak{W}_\tau$ contains all elements of $\mathfrak{B}$. Further, every element of $\mathfrak{W}_\tau$ already occurs in some system $\mathfrak{W}_\rho$ ($\rho < \tau$), and $\mathfrak{W}_\rho$ is equivalent to $\mathfrak{W}_0$; hence $\mathfrak{W}_\tau$ is algebraically dependent on $\mathfrak{W}_0$. To prove g) it remains only to show that $\mathfrak{W}_0$ is also algebraically dependent on $\mathfrak{W}_\tau$. For this purpose we give a more general proof.

We denote by $\mathfrak{W}_{\mu\nu}$ ($0 \leq \mu \leq \tau$, $0 < \nu < \sigma$) the segment of $\mathfrak{W}_\mu$ determined by $c_{\mu\nu}$, and put $\mathfrak{W}_\mu = \mathfrak{W}_{\mu\sigma}$. We further denote by $\mathfrak{W}'_{\mu\nu}$ ($0 \leq \mu \leq \tau$, $0 < \nu \leq \sigma$) that subsystem of $\mathfrak{W}_\mu$ which contains all elements of $\mathfrak{W}_{\mu\nu}$, all elements of $\mathfrak{B}_\mu$, and no other element; and put $\mathfrak{W}'_{0\nu} = \mathfrak{W}_{0\nu}$ ($0 < \nu \leq \sigma$). We now assert:

h) Every system $\mathfrak{W}'_{\mu\nu}$ ($0 \leq \mu \leq \tau$, $0 < \nu \leq \sigma$) is algebraically dependent on every system $\mathfrak{W}'_{\rho\nu}$ ($\mu \leq \rho < \tau$).

This assertion includes, when we choose $\mu = 0$, $\nu = \sigma$, $\rho = \tau$, the assertion that $\mathfrak{W}_0$ is algebraically dependent on $\mathfrak{W}_\tau$. Thus h) proves g), and hence e). If h) did not hold for every system $\mathfrak{W}'_{\mu\nu}$, we would choose among the systems for which it fails one for which the second

index ν is as small as possible. We then choose for this $\mathfrak{W}'_{\mu\nu}$ the system $\mathfrak{W}'_{\rho\nu}$ with least index $\rho > \mu$ on which $\mathfrak{W}'_{\mu\nu}$ is not algebraically dependent. Let a be an element of $\mathfrak{W}'_{\mu\nu}$ which is not algebraically dependent on $\mathfrak{W}'_{\rho\nu}$. Then a does not belong to $\mathfrak{W}'_{\rho\nu}$, hence not to $\mathfrak{W}_\rho$, and also not to $\mathfrak{B}_\rho$; consequently it belongs to $\mathfrak{W}_{\mu\nu}$. It must moreover be the last element of $\mathfrak{W}_{\mu\nu}$. Otherwise a would belong to a segment $\mathfrak{W}_{\mu\lambda}$ of $\mathfrak{W}_{\mu\nu}$, and hence also to the system $\mathfrak{W}'_{\mu\lambda}$; since this system, by the minimal property of ν , is algebraically dependent on $\mathfrak{W}'_{\rho\lambda}$, and $\mathfrak{W}'_{\rho\lambda}$ is a part of $\mathfrak{W}'_{\rho\nu}$, the element a would be algebraically dependent on $\mathfrak{W}'_{\rho\nu}$. Thus a is the last element of $\mathfrak{W}_{\mu\nu}$; hence ν is not a limit number, and

$$a = c_{\mu,\nu-1}, \quad \mathfrak{W}_{\mu\nu} = \mathfrak{W}_{\mu,\nu-1} + c_{\mu,\nu-1},$$

and, since a does not belong to $\mathfrak{B}_\mu$,

$$\mathfrak{W}'_{\mu\nu} = \mathfrak{W}'_{\mu,\nu-1} + c_{\mu,\nu-1},$$

where in the case $\nu = 1$ both $\mathfrak{W}_{\mu,\nu-1}$ and $\mathfrak{W}'_{\mu,\nu-1}$ denote the empty system. Let λ run through all numbers $\geq \mu$ and $< \rho$. By the minimal property of ρ , $\mathfrak{W}'_{\mu\nu}$ is algebraically dependent on every system $\mathfrak{W}'_{\lambda\nu}$; therefore none of these systems may be algebraically dependent on $\mathfrak{W}'_{\rho\nu}$. But, apart from $c_{\lambda,\nu-1}$, all elements of $\mathfrak{W}'_{\lambda\nu}$ belong to $\mathfrak{W}'_{\lambda,\nu-1}$, and hence are algebraically dependent on $\mathfrak{W}'_{\rho,\nu-1}$, and therefore on $\mathfrak{W}'_{\rho\nu}$. Consequently $c_{\lambda,\nu-1}$ cannot be algebraically dependent on $\mathfrak{W}'_{\rho\nu}$. Thus the elements $c_{\lambda,\nu-1}$ must be distinct from the element $c_{\rho,\nu-1}$ contained in $\mathfrak{W}'_{\rho\nu}$. It follows that ρ is not a limit number and

$$(1) \quad c_{\rho-1,\nu-1} = u_{\nu-1}, \quad c_{\rho,\nu-1} = v_{\rho-1}, \quad \mathfrak{W}_\rho = \vartheta_{\rho-1}(\mathfrak{W}_{\rho-1}, \mathfrak{B})$$

holds. Let $\mathfrak{L}$ denote the least subsystem of $\mathfrak{W}_{\rho-1}$ on which $v_{\rho-1}$ is algebraically dependent. By the definition of $\vartheta_{\rho-1}$, $\mathfrak{W}_\rho$ is obtained from $\mathfrak{W}_{\rho-1}$ by replacing the last element of $\mathfrak{L}$ not belonging to $\mathfrak{B}_{\rho-1}$ by $v_{\rho-1}$. In view of (1), this last element must be $c_{\rho-1,\nu-1} = u_{\nu-1}$; that is, every element of $\mathfrak{L}$ belongs either to $\mathfrak{W}_{\rho-1,\nu}$ or to $\mathfrak{B}_{\rho-1}$. If therefore $\mathfrak{L}'$ denotes the system into which $\mathfrak{L}$ is transformed by substituting $v_{\rho-1}$ for $u_{\nu-1}$, then every element of $\mathfrak{L}'$ belongs either to $\mathfrak{W}'_{\rho\nu}$ or to $\mathfrak{B}_{\rho-1}$; thus $\mathfrak{L}'$ is part of $\mathfrak{W}'_{\rho\nu}$. On the other hand $\mathfrak{L}'$ is equivalent to $\mathfrak{L}$, and the element $a = u_{\nu-1} = c_{\rho-1,\nu-1}$ contained in $\mathfrak{L}$ is therefore algebraically dependent on $\mathfrak{W}'_{\rho\nu}$. This contradicts the preceding conclusion. Hence the assertion h) is true for all systems $\mathfrak{W}'_{\mu\nu}$, and e) is proved.

For our purpose the only point of interest is that the system $\mathfrak{B}$, as part of $\mathfrak{W}_\tau$, is mapped one-to-one onto a part of $\mathfrak{U}$ in the similar relation existing between $\mathfrak{U} = \mathfrak{W}_0$ and $\mathfrak{W}_\tau$. If the systems $\mathfrak{U}$ and $\mathfrak{B}$ are equivalent, then in the same way there is a one-to-one mapping of $\mathfrak{U}$ onto a part of $\mathfrak{B}$. By a known theorem of Bernstein it follows that there is also a one-to-one correspondence between $\mathfrak{U}$ and $\mathfrak{B}$, and hence:

3. Equivalent irreducible systems have the same cardinal number.

When a transcendental extension Ω of a field $\mathfrak{K}$ is decomposed, by inserting a field $\mathfrak{K}(\mathfrak{S})$, into a purely transcendental extension followed by an algebraic one, all possible such decompositions are obtained from this one by replacing the irreducible system $\mathfrak{S}$ by all possible equivalent irreducible systems. The cardinal number of $\mathfrak{S}$ is therefore completely determined by $\mathfrak{K}$ and Ω . We call it the *transcendence degree of the extension*, or the *transcendence degree of Ω with respect to $\mathfrak{K}$* . We assign transcendence degree 0 to algebraic extensions.

Let Ω be a field between $\mathfrak{K}$ and $\mathfrak{M}$, let $\mathfrak{S}$ be a system of elements of Ω irreducible with respect to $\mathfrak{K}$, and let $\mathfrak{T}$ be a system of elements of $\mathfrak{M}$ irreducible with respect to Ω . Suppose Ω is algebraic with respect to $\mathfrak{K}(\mathfrak{S})$, and $\mathfrak{M}$ algebraic with respect to $\Omega(\mathfrak{T})$, so that the cardinal numbers s and t of $\mathfrak{S}$ and $\mathfrak{T}$ are the transcendence degrees of Ω with respect to $\mathfrak{K}$, and of $\mathfrak{M}$ with respect to Ω . Then, with respect to $\mathfrak{K}(\mathfrak{S} + \mathfrak{T})$, all elements of Ω and all elements of $\mathfrak{T}$ are algebraic. But by adjoining all these elements to $\mathfrak{K}(\mathfrak{S} + \mathfrak{T})$ one obtains the field $\Omega(\mathfrak{T})$, which

is therefore algebraic with respect to $\mathfrak{K}(\mathfrak{S} + \mathfrak{T})$. Since likewise $\mathfrak{M}$ is algebraic with respect to $\Omega(\mathfrak{T})$, $\mathfrak{M}$ is an algebraic extension of $\mathfrak{K}(\mathfrak{S} + \mathfrak{T})$.

Consider an arbitrary finite subsystem of $\mathfrak{S} + \mathfrak{T}$, and write it in the form $\mathfrak{S}' + \mathfrak{T}'$, where $\mathfrak{S}'$ is part of $\mathfrak{S}$ and $\mathfrak{T}'$ part of $\mathfrak{T}$. The system $\mathfrak{S}'$ is irreducible with respect to $\mathfrak{K}$, and $\mathfrak{T}'$ is irreducible with respect to Ω , hence also with respect to $\mathfrak{K}(\mathfrak{S}')$. If therefore $\mathfrak{S}'$ consists of the elements $x_1, \dots, x_m$, and $\mathfrak{T}'$ of the elements $y_1, \dots, y_n$, then of the elements $x_1, \dots, x_m, y_1, \dots, y_n$, each is transcendental over the field obtained by adjoining the preceding ones to $\mathfrak{K}$. Thus, with respect to $\mathfrak{K}$, none of these elements is algebraically dependent on the system of the remaining ones. Hence the system $\mathfrak{S}' + \mathfrak{T}'$ is irreducible with respect to $\mathfrak{K}$; since it was an arbitrary finite subsystem of $\mathfrak{S} + \mathfrak{T}$, the system $\mathfrak{S} + \mathfrak{T}$ itself is irreducible with respect to $\mathfrak{K}$ (§ 22, 4). The cardinal number of $\mathfrak{S} + \mathfrak{T}$ is denoted by $s + t$, so that the following theorem holds:

4. *An extension composed of two successive extensions of transcendence degrees s and t has transcendence degree $s + t$.*

It is immediate that this remains valid when s or t is 0.

By the *absolute transcendence degree* of a field $\mathfrak{K}$ we mean its transcendence degree over the prime field; and we collect in a class $K_{c,t}$ all fields (or field types) which have the same characteristic c and the same absolute transcendence degree t . Here c may be 0 or any prime number, and t may be 0, a natural number, or any transfinite cardinal number. In each class $K_{c,t}$ there are two notable types. Namely, if one takes a prime field $\mathfrak{P}_c$ and forms a purely transcendental extension of transcendence degree t , one reaches a well-determined type (§ 22, 5), which belongs to the class $K_{c,t}$ and will be denoted by $\mathfrak{R}_{c,t}$. To this type there further belongs, by § 21, 9, a definite algebraically closed type; call it $\mathfrak{A}_{c,t}$. It arises from $\mathfrak{R}_{c,t}$ by algebraic extension, and therefore also belongs to the class $K_{c,t}$.

5. *If a field $\mathfrak{R}'_{c,t}$ of type $\mathfrak{R}_{c,t}$ is algebraically extended to an algebraically closed field $\mathfrak{A}'_{c,t}$, then all fields between $\mathfrak{R}'_{c,t}$ and $\mathfrak{A}'_{c,t}$ belong to the class $K_{c,t}$. Moreover every type of the class $K_{c,t}$ is represented by at least one field between $\mathfrak{R}'_{c,t}$ and $\mathfrak{A}'_{c,t}$.*

That all fields between $\mathfrak{R}'_{c,t}$ and $\mathfrak{A}'_{c,t}$ belong to the class $K_{c,t}$ follows from the fact that they are algebraic with respect to $\mathfrak{R}'_{c,t}$, and that $\mathfrak{R}'_{c,t}$ belongs to this class. Next it follows from our hypotheses that $\mathfrak{A}'_{c,t}$ is of type $\mathfrak{A}_{c,t}$. Let now $\mathfrak{K}$ be any field of the class $K_{c,t}$, and let $\mathfrak{P}_c$ be the prime field contained in $\mathfrak{K}$. We may choose from $\mathfrak{K}$ a system $\mathfrak{S}$, irreducible with respect to $\mathfrak{P}_c$, such that $\mathfrak{K}$ is algebraic with respect to $\overline{\mathfrak{R}}_{c,t} = \mathfrak{P}_c(\mathfrak{S})$. If we algebraically extend $\mathfrak{K}$ to an algebraically closed field $\overline{\mathfrak{A}}_{c,t}$, then this field is also algebraic with respect to $\overline{\mathfrak{R}}_{c,t}$. The fact that $\mathfrak{K}$ belongs to $K_{c,t}$ implies that $\mathfrak{S}$ has cardinal number t , that $\overline{\mathfrak{R}}_{c,t}$ is of type $\mathfrak{R}_{c,t}$, and that $\overline{\mathfrak{A}}_{c,t}$ is of type $\mathfrak{A}_{c,t}$. It follows further (compare the proof of § 21, 9) that $\overline{\mathfrak{A}}_{c,t}$ can be mapped isomorphically onto $\mathfrak{A}'_{c,t}$ in such a way that $\overline{\mathfrak{R}}_{c,t}$ passes into $\mathfrak{R}'_{c,t}$. Under this mapping $\mathfrak{K}$ is placed in isomorphic correspondence with a field between $\mathfrak{R}'_{c,t}$ and $\mathfrak{A}'_{c,t}$; hence the type of $\mathfrak{K}$ is represented among these fields. If $\mathfrak{K}$ itself is algebraically closed, then $\mathfrak{K} = \overline{\mathfrak{A}}_{c,t}$ is of type $\mathfrak{A}_{c,t}$. Therefore:

6. *The type $\mathfrak{A}_{c,t}$ is the only algebraically closed type in the class $K_{c,t}$.*

§ 24. Additional Remarks of Various Kinds.

The Cardinal Number (Power, Number of Elements) of a Field. – From the well-ordering theorem it follows easily that cardinal numbers also form a well-ordered sequence; that the sum and the product of two cardinal numbers, at least one of which is transfinite, is equal to the larger one; and that the system of all finite subsets of an infinite set M has the same power as M . From this one further concludes that the number of integral functions in one indeterminate x over an infinite field is no greater than the number of elements of the field, and that the same is true for the roots of these functions. Algebraic extension of an infinite field therefore does

not increase its power. Another simple consequence of the cited theorems is that, in the case $t \geq \aleph_0$ (where $\aleph_0$ denotes the least transfinite cardinal number, the power of the countable sets), the field $\mathfrak{K}_{c,t}$, and hence every field of the class $K_{c,t}$, has cardinal number t ; while in the case of finite t , a field of the class $K_{c,t}$, if it is not finite (which can occur only for $c = p$, $t = 0$), has $\aleph_0$ elements. Thus for a field whose elements are not countable, the cardinal number agrees with the absolute transcendence degree. Since the continuum is not countable, it follows (§ 23, 6) that all algebraically closed fields of the power of the continuum and of characteristic 0 are isomorphic. Among these fields are: the field of complex numbers, the field of all algebraic functions of one or more variables, and others.

Lüroth's Theorem. – Beside Theorem II of § 14 stands the following:

If $\mathfrak{K}(x)$ is a simple transcendental extension of $\mathfrak{K}$, then every field between $\mathfrak{K}$ and $\mathfrak{K}(x)$ is also simple with respect to $\mathfrak{K}$.

We can also formulate this theorem as follows: if Ω is a purely transcendental extension of $\mathfrak{K}$ of transcendence degree 1, then every field between $\mathfrak{K}$ and Ω is likewise a purely transcendental extension of $\mathfrak{K}$. The natural question whether the same theorem holds for every purely transcendental extension has not yet been settled.

To arrive at a proof of Lüroth's theorem, introduce beside x a new indeterminate t , and consider the integral functions of t with coefficients in $\mathfrak{K}(x)$. To every such function $g(t)$ there belongs a function obtained from it by multiplication by a factor from $\mathfrak{K}(x)$, primitive with respect to x ,

$$\bar{g}(t) = \varphi_0(x)t^n + \varphi_1(x)t^{n-1} + \cdots + \varphi_n(x),$$

which is distinguished by the fact that its coefficients are integral functions of x not all having a common divisor. This function is determined up to a factor from $\mathfrak{K}$; if one wants it to be completely determined, one may further require that in $\varphi_0(x)$ the coefficient of the highest power shall be ε . It is shown in the usual way that the product of two primitive functions is again primitive; from this it follows immediately that from an equation of the form $f(t) = g(t)h(t)$, where $f(t), g(t), h(t)$ are integral functions of t with coefficients in $\mathfrak{K}(x)$, one again obtains a correct equation by replacing these functions by the associated primitive ones.

Let $\varphi(t)/\psi(t)$ be a rational function of t , with coefficients in $\mathfrak{K}$, in reduced form; let m be its degree (> 0), and let y be an element transcendental with respect to $\mathfrak{K}(t)$. Then the function

$$\varphi(t) - y\psi(t)$$

has degree m in t ; it is primitive in y , and, as we shall show at once, irreducible in the field $\mathfrak{K}(y)$. For if there were a decomposition

$$\varphi(t) - y\psi(t) = g(t)h(t),$$

then, on passing to the functions primitive in y , the left hand side would not change; hence we may assume at once that $g(t)$ and $h(t)$ are primitive. One of these functions, say $g(t)$, would then have to be free of y , and the other of first degree in y . Ordering by powers of y , one sees that $g(t)$ would have to divide both $\varphi(t)$ and $\psi(t)$, whereas $\varphi(t)$ and $\psi(t)$ were assumed relatively prime. Put

$$y = \frac{\varphi(x)}{\psi(x)}.$$

Then $\mathfrak{K}(y)$ is a field between $\mathfrak{K}$ and $\mathfrak{K}(x)$, and x is a root of $\varphi(t) - y\psi(t)$, which has degree m and is irreducible over $\mathfrak{K}(y)$. Hence $[\mathfrak{K}(x) : \mathfrak{K}(y)] = m$, i.e. the degree of $\mathfrak{K}(x)$ with respect to $\mathfrak{K}(y)$ is equal to the degree of y as a rational function of x .

We now show that the function

$$\psi(x)(\varphi(t) - y\psi(t)) = \varphi(t)\psi(x) - \psi(t)\varphi(x)$$

is primitive in x as an integral function of t . For this purpose put

$$\varphi(x) = a_0 + a_1x + \cdots, \quad \psi(x) = b_0 + b_1x + \cdots,$$

so that

$$\varphi(t)\psi(x) - \psi(t)\varphi(x) = (a_0\psi(x) - b_0\varphi(x)) + (a_1\psi(x) - b_1\varphi(x))t + \cdots.$$

If all coefficients $\gamma_i = a_i\psi(x) - b_i\varphi(x)$ had a common factor $\chi(x)$, then this factor would also divide all expressions

$$\gamma_i b_k - \gamma_k b_i = (a_i b_k - a_k b_i)\psi(x), \quad \gamma_i a_k - \gamma_k a_i = (a_i b_k - a_k b_i)\varphi(x),$$

and hence, since (as $m > 0$) we can choose i and k so that $a_i b_k - a_k b_i \neq 0$, it would also divide $\varphi(x)$ and $\psi(x)$, contradicting the assumption that $\varphi(x)$ and $\psi(x)$ are relatively prime. Therefore, if $\varphi(t)\psi(x) - \psi(t)\varphi(x)$ decomposes in the integral domain $\mathfrak{K}[t, x]$ into two factors, neither of them can be a function of x alone. In precisely the same way one shows that neither of the two factors can be a function of t alone.

Let now $\mathfrak{M}$ be an arbitrary field, distinct from $\mathfrak{K}$, between $\mathfrak{K}$ and $\mathfrak{K}(x)$. Then $\mathfrak{K}(x)$ is finite over $\mathfrak{M}$; for if y is an element of $\mathfrak{M}$ not contained in $\mathfrak{K}$, then, as we saw, $\mathfrak{K}(x)$ is finite over $\mathfrak{K}(y)$, and $\mathfrak{M}$ lies between $\mathfrak{K}(y)$ and $\mathfrak{K}(x)$. Let

$$(1) \quad [\mathfrak{K}(x) : \mathfrak{M}] = n.$$

Then x is a root of an irreducible function of degree n over $\mathfrak{M}$,

$$g(t) = t^n + u_1 t^{n-1} + \cdots + u_n,$$

whose coefficients, since x is transcendental with respect to $\mathfrak{K}$, cannot all occur in $\mathfrak{K}$. Let

$$y = \frac{\varphi(x)}{\psi(x)}$$

in reduced form be one of the coefficients of $g(t)$ which does not occur in $\mathfrak{K}$, and let m be its degree in x . Then $\mathfrak{M}$ lies between $\mathfrak{K}(y)$ and $\mathfrak{K}(x)$, and we obtain $[\mathfrak{K}(x) : \mathfrak{K}(y)] = m$, hence, by (1),

$$(2) \quad [\mathfrak{M} : \mathfrak{K}(y)] = \frac{m}{n}, \quad m \geq n.$$

To $g(t)$ there belongs a function primitive in x ,

$$\bar{g}(t) = \varphi_0(x)t^n + \varphi_1(x)t^{n-1} + \cdots + \varphi_n(x),$$

where

$$\frac{\varphi_k(x)}{\varphi_0(x)} = u_k \quad (k = 1, \dots, n).$$

Since among the functions u_k there also occurs y , $\bar{g}(t)$ has degree at least m in x . Now x is a root of the function $\varphi(t) - y\psi(t)$ belonging to the field $\mathfrak{M}$; this function must therefore be divisible by the function $g(t)$ irreducible over $\mathfrak{M}$. Thus we get

$$\varphi(t) - \frac{\varphi(x)}{\psi(x)}\psi(t) = g(t)h(t),$$

and, on passing to the associated functions primitive in x ,

$$(3) \quad \varphi(t)\psi(x) - \psi(t)\varphi(x) = \bar{g}(t)\bar{h}(t).$$

The left hand side of (3) has degree m in x ; $\bar{g}(t)$, as we saw, has degree at least m in x . It follows from (3) that $\bar{g}(t)$ has exactly degree m in x , and that $\bar{h}(t)$ is free of x . Since it was

further shown that $\varphi(t)\psi(x) - \psi(t)\varphi(x)$ cannot have a factor that is a function of t alone, $\bar{h}(t)$ cannot contain t either. Comparing degrees in t on both sides of (3), we obtain

$$m = n,$$

and from (2)

$$\mathfrak{M} = \mathfrak{K}(y).$$

This proves Lüroth's theorem.

This “rational” proof of the theorem has the advantage that it holds without any restriction for all fields $\mathfrak{K}$, whereas the usual form of the proof contains the assumptions that the field $\mathfrak{K}$ is perfect and has infinitely many elements. In the case of an imperfect field that form of proof requires cumbersome modifications; for finite fields it fails altogether.

The Algebraically Closed Extensions. – Proving the existence of an algebraically closed extension belonging to a given field $\mathfrak{K}$ still gives no method for actually constructing such an extension. Even for arranging the elements of a field $\mathfrak{K}$ into a well-ordered system, to which that proof refers, there is naturally no general method. The construction of an algebraically closed extension is, however, by no means tied to an already carried-out well-ordering of $\mathfrak{K}$. Two examples follow. In both, the ground field has the power of the continuum, of which, as is known, it has not even been established whether it is equal to $\aleph_1$, the second transfinite cardinal number. Of this power are, among others, the field of real or complex numbers, the field of rational functions of one variable z with complex coefficients, and also the field $K(p)$ of the so-called p -adic numbers introduced by K. Hensel.¹

In the case $\mathfrak{K} = K(p)$, one reaches the goal on the basis of various investigations which Hensel carried out in the cited work and in a paper recently appearing in this journal.² The essential point is the proof that every integral function $f(x)$ over $K(p)$ splits completely in a field obtained from $K(p)$ by adjoining a root of a function $g(x)$ which is irreducible over $K(p)$ and whose coefficients are ordinary integers. Thus it is proved that the system of all integral functions over $K(p)$ is equivalent to the system S of integral functions with integral coefficients. While the first-mentioned system has the power of the continuum, the system S is countable and may therefore be arranged in a simple sequence. Once one has decided on some order and denotes the functions so ordered by $f_n(x)$, where n runs through the natural numbers, Hensel's methods allow one, by a finite procedure, to compute the first function from S which is equivalent to the system

$$f_1(x), f_2(x), \dots, f_n(x),$$

irreducible over $K(p)$, and moreover normal. If this function is denoted by $\varphi_n(x)$, one has a system of normal functions

$$\varphi_1(x), \varphi_2(x), \dots,$$

in which each segment is subordinated to the following function. Introducing a system of symbols j_n , with the stipulation that j_n is to be a root of $\varphi_n(x)$, one obtains a system of extensions of the field $\mathfrak{K} = K(p)$:

$$(4) \quad \mathfrak{K}_1, \mathfrak{K}_2, \dots$$

Let r_n denote the degree of $\varphi_n(x)$. Then every element of $\mathfrak{K}_n$ can be represented in the form $g(j_n)$, where $g(x)$ is an integral function over $K(p)$ of degree less than r_n . Among the elements of $\mathfrak{K}_n$, r_{n-1} are roots of $\varphi_{n-1}(x)$. If we want every field of the sequence (4) to be part of the following one, so that, as in the case treated in § 16, these fields together constitute the desired algebraically closed field, we must still make a stipulation by which it is decided which of the roots of $\varphi_{n-1}(x)$ contained in $\mathfrak{K}_n$ is to be equal to j_{n-1} . Now it has indeed not yet been possible

¹Theorie der algebraischen Zahlen I (Leipzig and Berlin 1908).

²Vol. 136 (1909), p. 183.

to represent $K(p)$, or any system of the power of the continuum, as a well-ordered system; but $K(p)$ can very easily be represented as an ordered system.³ On the basis of such an order of $K(p)$, one obtains an order of the integral functions over $K(p)$ by stipulating that the function $g(x) = a_0 + a_1x + \cdots$ shall precede the function $h(x) = b_0 + b_1x + \cdots$ when, in the case $a_k = b_k$ ($k < n$), $a_n \neq b_n$, the element a_n is earlier than b_n . One can then require that, if $g(x)$ is the first function over $K(p)$ which, for $x = j_n$, goes over into a root of $\varphi_{n-1}(x)$, then $g(j_n) = j_{n-1}$. This solves the problem.

As a second case consider the field $\mathfrak{K} = \mathbb{C}(z)$, obtained from the field $\mathbb{C}$ of complex numbers by adjoining a transcendental element z (the field of rational functions in one variable). Here the system of all integral functions of one indeterminate x over $\mathfrak{K}$ is no longer equivalent to a countable system; yet the construction of an algebraically closed extension of $\mathfrak{K}$ (without the principle of choice) is easily achieved using the means of analysis. Namely, let $\mathfrak{A}$ be the system of all series, convergent for sufficiently small values of r , of the form

$$\sum_{k=q}^{+\infty} a_k r^{k/n} \left(\cos \frac{k}{n} \varphi + i \sin \frac{k}{n} \varphi \right),$$

where r is a positive variable, φ a real variable, n an arbitrary natural number, q an arbitrary integer, $r^{k/n}$ is always to be taken positive, and the coefficients a_k , whose index k runs through all integers from q upward, are arbitrary complex numbers. Each such series represents, in its region of convergence, a single-valued function of r and φ . Two series are regarded as the same element of $\mathfrak{A}$ if, for all pairs of values r, φ in their common region of convergence, they converge to the same value. To two elements α, β of $\mathfrak{A}$ there belong, as is shown in the usual way, two further elements γ, δ , such that within the common convergence region the relations between $\alpha, \beta, \gamma, \delta$, regarded as functions of r and φ , are

$$(5) \quad \alpha + \beta = \gamma, \quad \alpha\beta = \delta.$$

If we let equations (5) serve as composition equations for $\mathfrak{A}$, then $\mathfrak{A}$ becomes a system with double composition, which is easily seen to be an algebraically closed field. To each element $R(z)$ of $\mathfrak{K} = \mathbb{C}(z)$ there corresponds in $\mathfrak{A}$ an element α such that, for sufficiently small values of r , $R(r(\cos \varphi + i \sin \varphi))$ becomes equal to α . The field $\mathfrak{K}$ is thus mapped isomorphically onto a subfield $\mathfrak{K}'$ of $\mathfrak{A}$; and since $\mathfrak{A}$ is algebraically closed, the elements of $\mathfrak{A}$ algebraic with respect to $\mathfrak{K}'$ form a field $\mathfrak{A}'$, which is algebraically closed and algebraic with respect to $\mathfrak{K}'$.

Isomorphic Mapping of a Field onto a Subfield. – Let Ω be an algebraic extension of the field $\mathfrak{K}$, let Ω' be a field between $\mathfrak{K}$ and Ω and different from Ω , let α be an element of Ω not contained in Ω' , and let $\varphi(x)$ be the irreducible function over $\mathfrak{K}$ with root α . Then the number of irreducible factors of $\varphi(x)$ in the field Ω is greater than in the field Ω' ; hence Ω and Ω' are not equivalent extensions. If $\mathfrak{K}$ is a prime field, then Ω and Ω' cannot be isomorphic, because in an isomorphic relation between Ω and Ω' every element of the prime field would remain unchanged, and Ω and Ω' would therefore be equivalent extensions of $\mathfrak{K}$. Thus:

1. *An absolutely algebraic field is not isomorphic to any proper part of itself.*

Since every subfield of an absolutely algebraic field is again absolutely algebraic, we have:

2. *Two subfields of an absolutely algebraic field, one of which is a proper part of the other, cannot be isomorphic.*

The property described in 2 is characteristic of absolutely algebraic fields. For if the field $\mathfrak{K}$ is not absolutely algebraic and x is transcendental with respect to its prime field $\mathfrak{P}$, then $\mathfrak{P}(x^2)$ is a proper part of $\mathfrak{P}(x)$, but the two fields are isomorphic. In contrast, 1 expresses no characteristic property of absolutely algebraic fields; for one sees very easily, for example, that a field of type $\mathfrak{A}_{c,t}$, where t is finite, cannot be isomorphic to any proper part of itself.

³Hensel, Neue Grundlagen der Arithmetik. This journal, Vol. 127, pp. 52, 53.

Among the fields that are isomorphic to a proper part of themselves, we have to distinguish two kinds. A representative of the first kind is the field $\mathfrak{R}_{c,1}$. It possesses infinitely many subfields, but, except for the prime field, all are of type $\mathfrak{R}_{c,1}$ by Lüroth's theorem. A field lying between a field $\mathfrak{R}_{c,1}$ and an isomorphic subfield is therefore always again isomorphic to $\mathfrak{R}_{c,1}$. Of another kind is the field $\mathfrak{A}_{c,t}$ for infinite t . By adjoining one transcendental element x , from $\mathfrak{A}_{c,t}$ one obtains a field $\mathfrak{A}_{c,t}(x)$ which again has transcendence degree t , but is plainly not algebraically closed, and so is not of type $\mathfrak{A}_{c,t}$. If, however, we algebraically extend $\mathfrak{A}_{c,t}(x)$ to an algebraically closed field, this field is again of type $\mathfrak{A}_{c,t}$. Here, therefore, between two isomorphic fields lies a non-isomorphic one. Whenever such a case occurs, we have two fields Ω_1, Ω_2 of different types, each of which is isomorphic to a part of the other. The subfields of Ω_1 therefore represent the same types as the subfields of Ω_2 .

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